Assignment 3: AWS Architecture Report

AWS Architecture Report

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42904 Cloud Computing and Software as A Service

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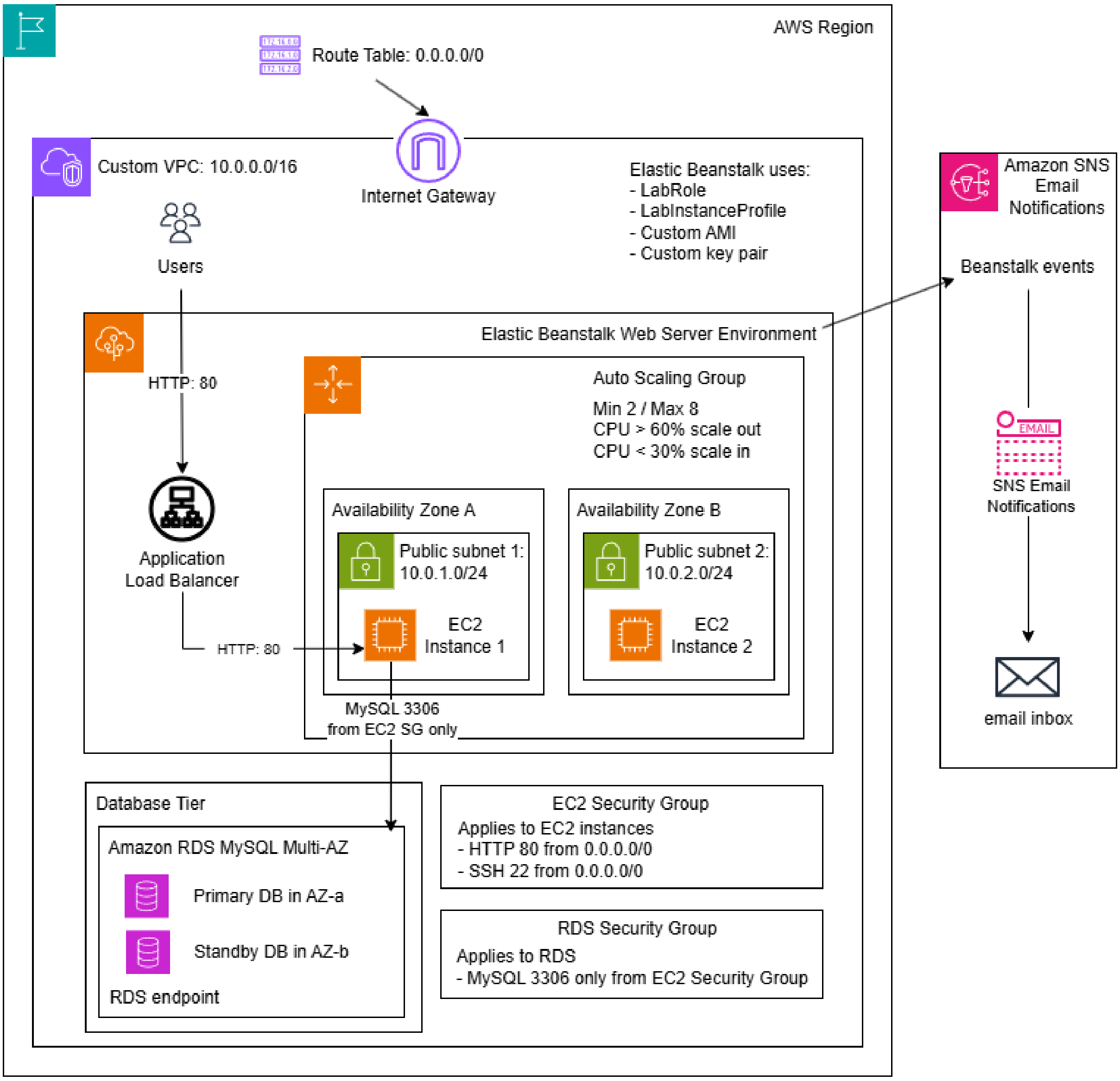
1 Overview of the architecture

1.1 Architecture diagram



1.2 Justification for each AWS component

AWS Elastic Beanstalk: The application uses Elastic Beanstalk to deploy and manage FreshBasket website. It works with EC2 instances, the Auto Scaling Group, and the Application Load Balancer. This reduces the manual setup workload. It also allows the environment to use a custom VPC, custom security groups, a custom AMI, and a custom key pair.

Application Load Balancer: The Application Load Balancer is the public entry point for user HTTP requests. It receives traffic from users on port 80. It then sends the traffic to EC2 instances in different Availability Zones. This supports scalability because traffic can be shared between multiple instances. It also supports fault tolerance because the load balancer can send traffic to healthy instances if one instance fails.

Auto Scaling Group: The Auto Scaling Group manages the number of EC2 instances for the application. It uses a minimum of 2 instances and a maximum of 8 instances. It adds more instances when CPU utilisation is above 60%. It removes instances when CPU utilisation is below 30%. This helps the system cope with busy periods while only running core servers during quiet periods.

Amazon EC2 instances: Amazon EC2 instances run the PHP web application. Elastic Beanstalk launches these instances inside the Auto Scaling Group. The design places at least two EC2 instances across two public subnets in different Availability Zones. This helps the application keep running if one instance or one Availability Zone has a problem.

Custom AMI: The custom AMI is created from a configured EC2 builder instance. It is used as the base image for the Elastic Beanstalk EC2 instances. This helps each new EC2 instance use the same server setup. It also supports scaling because the Auto Scaling Group can launch more instances from the same prepared image when traffic increases.

Custom key pair: The custom key pair is used for SSH access to the EC2 instances. It provides controlled administrative access and avoids relying on a default key pair.

Custom VPC: The custom VPC provides an isolated network environment for the FreshBasket application. It gives control over IP address ranges, subnets, routing, and security groups. This supports a clearer and more controlled production-style architecture.

Public subnets in two Availability Zones: The design uses two public subnets in two different Availability Zones. This supports high availability and disaster recovery at the Availability Zone level. If one Availability Zone fails, the other Availability Zone can still run the application. The design uses at least two instances across two Availability Zones, but more instances can be added by Auto Scaling.

Internet Gateway and route table: The Internet Gateway allows resources in the public subnets to connect to the internet. The route table includes a default route, 0.0.0.0/0, to the Internet Gateway. This allows users on the internet to reach the Application Load Balancer by using HTTP.

EC2 Security Group: The EC2 security group is applied to the EC2 application instances. It allows inbound HTTP traffic on port 80 so users can access the web application. It also allows SSH traffic on port 22 for administration and testing. This follows the assignment requirement to use one custom EC2 security group for the Auto Scaling instances.

RDS Security Group: The RDS security group is applied to the database. It only allows MySQL traffic on port 3306 from the EC2 security group. This protects the database because users on the public internet cannot access it directly.

Amazon RDS MySQL Multi-AZ: Amazon RDS MySQL stores the application data, such as FreshBasket order records. The Multi-AZ setup uses a primary database in one Availability

Zone and a standby database in another Availability Zone. This supports disaster recovery because the database can fail over if the primary database has a problem. The application can read from and write to RDS. This shows that the web tier and database are connected.

Amazon SNS email notifications: Email notifications are used for important Elastic Beanstalk environment events. The administrator can receive alerts when the environment changes state or has important events. This supports operational visibility of the system.

1.3 Statement of assumptions

This design is based on the assumption that FreshBasket is still a small start-up and that a majority of their users are in Australia. The application, therefore, is deployed in one AWS Region instead of multiple Regions. This helps making the first cloud deployment simple and cost conscious. The architecture also employs two Availability Zones within the Region, meaning that the application isn't reliant on one data centre. When one Availability Zone fails, the other one will continue to support the web application.

The application is also targeted at a small but increasing user base. Currently, about 200 people use FreshBasket every day, and could hit the count fast after gaining media attention. Due to this, the architecture begins with small EC2 instances like t3.micro, so as not to pay for unused capacity. When CPU utilisation increases, the Auto Scaling Group can add more EC2 instances. This supports traffic growth without requiring the startup to prepare many servers in advance. This is beneficial to traffic growth without the startup having to prepare a large number of servers ahead of time.

The Web application itself is kept as simple as possible as the central aim of this deployment is to illustrate the AWS architecture. Users can see the latest orders in the RDS MySQL database and submit a sample order for FreshBasket in the PHP page. That's sufficient for it to be able to go through the Load Balancer, to an EC2 instance in the Auto Scaling Group, and to read from or write to the RDS database.

The app uses port 80 for network access. In this first deployment HTTPS, custom domain and SSL certificates are not part of the main scope of the assignment. But if FreshBasket makes this system production ready, then they should be added. For testing and administration, SSH access is allowed on port 22. In a real production environment, SSH should be restricted to trusted IP addresses only. To have SSH access only for trusted IP addresses, in a real production environment it should be restricted to trusted IPs only.

A separate security group is used to protect the database tier. The RDS MySQL database cannot be accessed directly from the Internet. Rather, it will only allow MySQL traffic from the EC2 security group into port 3306 on the database. This implies that the application user has to access the application via the web layer and only the application instances can communicate with the database. This design is a simple separation of application tier and database tier.

Finally, disaster recovery is not Region-wide – it is Availability Zone level. This is appropriate for the size of a first cloud deployment. The solution uses a pair of public subnets across two AZs with at least two EC2 instances and RDS Multi-AZ. These decisions are a good fit for a small startup in terms of availability, scalability, and cost.

Table 1 List of every AWS service used

Service	Solution
AWS Elastic Beanstalk	To deploy and manage the PHP web application. It works with EC2 instances, the Auto Scaling Group, and the Application Load Balancer.
Amazon EC2	To run the application instances. It is also used to create the configured builder instance for the custom AMI.
Amazon Machine Image (AMI)	Created from the configured EC2 builder instance. Elastic Beanstalk uses this AMI as the base image for new application instances.
EC2 Key Pair	Using for SSH access to the EC2 instances.
Elastic Load Balancing	The public entry point for HTTP requests. It distributes traffic across the EC2 instances.
Amazon EC2 Auto Scaling	To manage the number of EC2 instances. It uses a minimum of 2 instances and a maximum of 8 instances. It supports CPU utilisation to decide when to scale out or scale in.
Amazon VPC	Providing the custom network environment, including the VPC, public subnets, route table, and Internet Gateway.
EC2 Security Groups	To control HTTP and SSH access to the application instances.
RDS Security Groups	The RDS security group controls MySQL access to the database.
Amazon RDS for MySQL	To store the application data. It is deployed in Multi-AZ mode to improve availability and support disaster recovery.
Amazon SNS	To send email notifications about Elastic Beanstalk environment events.
Amazon CloudWatch	To provide monitoring metrics used by Elastic Beanstalk and Auto Scaling. The Auto Scaling configuration uses CPUUtilization to scale out above 60% and scale in below 30%, while Elastic Beanstalk uses monitoring data to show environment health.
AWS IAM	To allow Elastic Beanstalk and EC2 to operate with the correct permissions in the Learner Lab environment. LabRole and LabInstanceProfile are reused to avoid permission errors from creating custom IAM roles.

2 The developed system in AWS

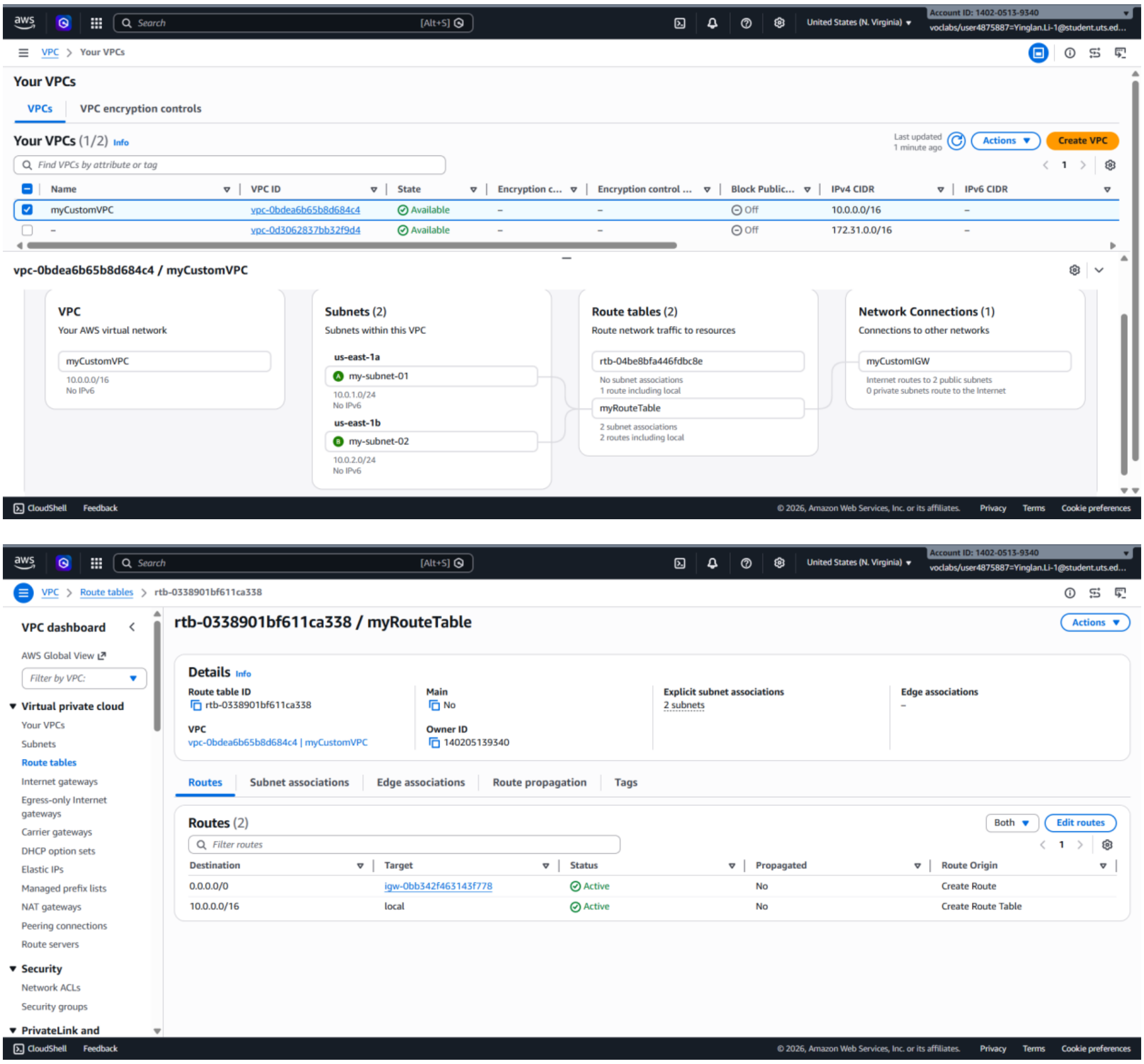


Figure 1. Custom VPC and public subnets

This screenshot shows the custom VPC, two public subnets in different Availability Zones, and the route table with a default route to the Internet Gateway.

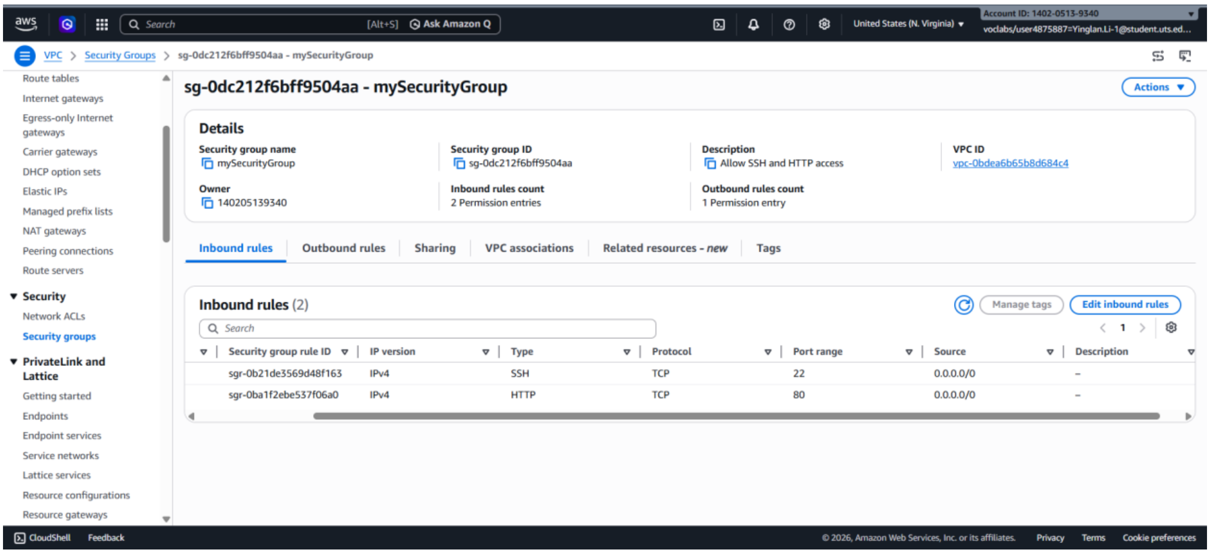


Figure 2. EC2 security group inbound rules

This screenshot shows the custom EC2 security group allowing inbound SSH traffic on port 22 and HTTP traffic on port 80.

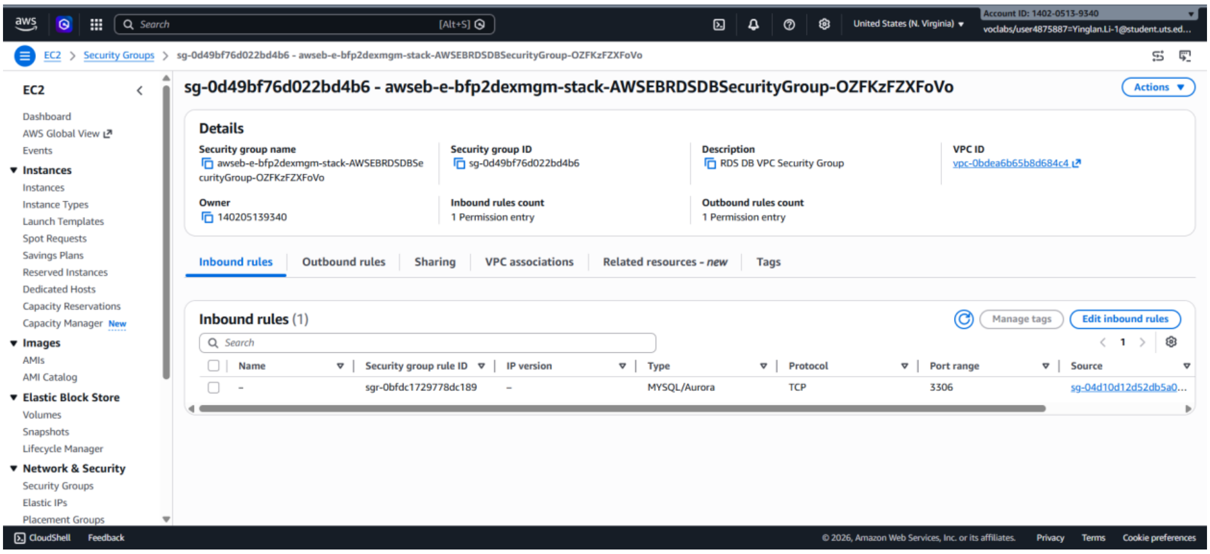


Figure 3. RDS security group inbound rule

This screenshot shows the RDS security group allowing MySQL/Aurora traffic on port 3306 only from the EC2 security group used by the Elastic Beanstalk instances.

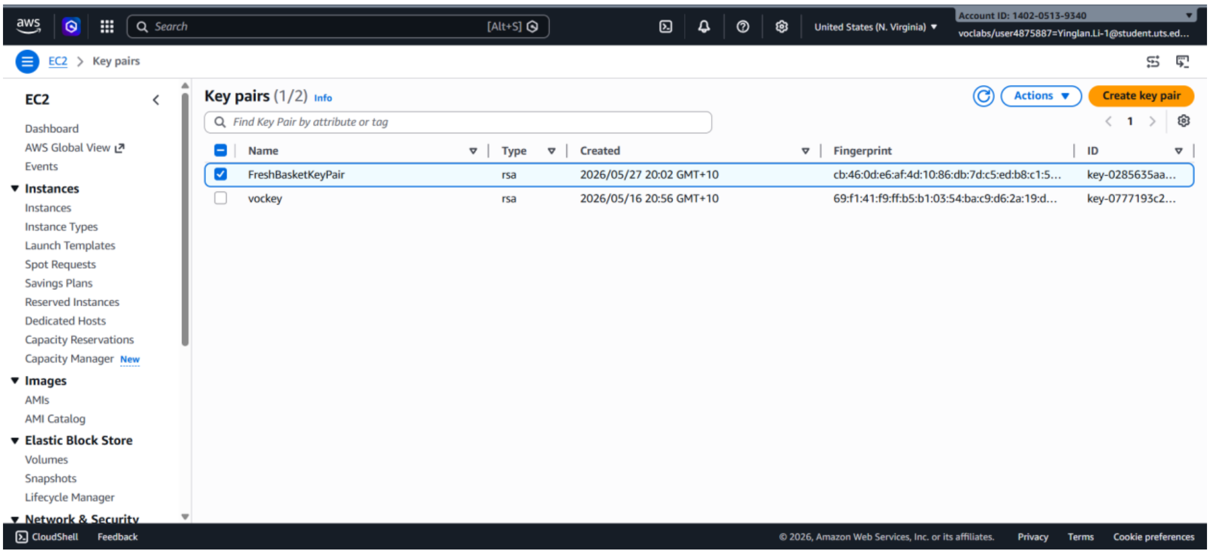


Figure 4. Custom key pair

This screenshot shows the custom key pair created for SSH access to the EC2 instances used in the deployment.

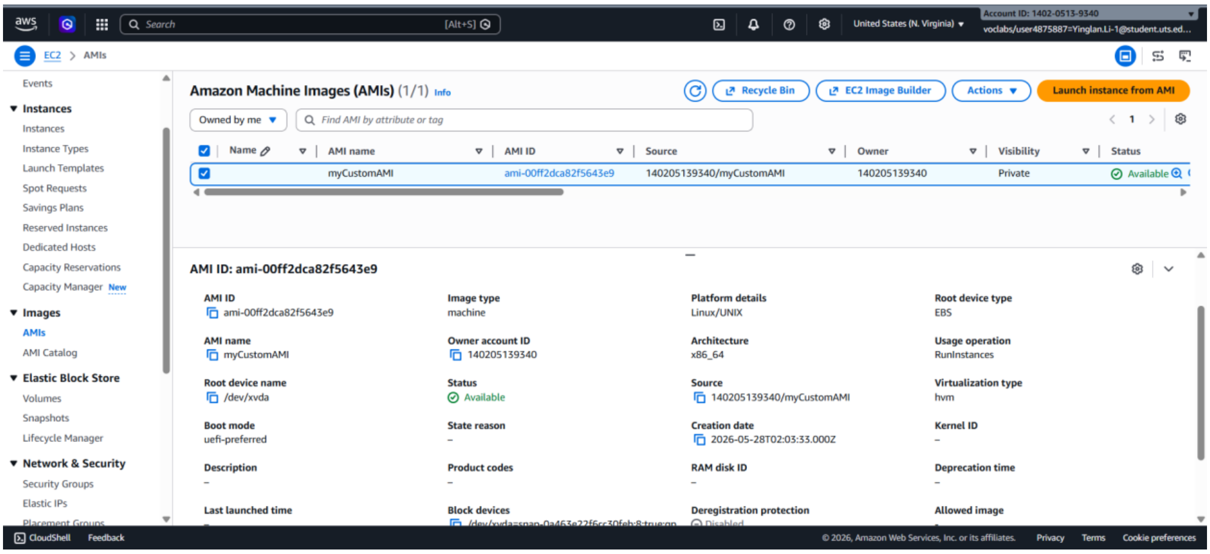


Figure 5. Custom AMI

This screenshot shows the custom AMI created from a configured EC2 builder instance and prepared for use by the Elastic Beanstalk environment.

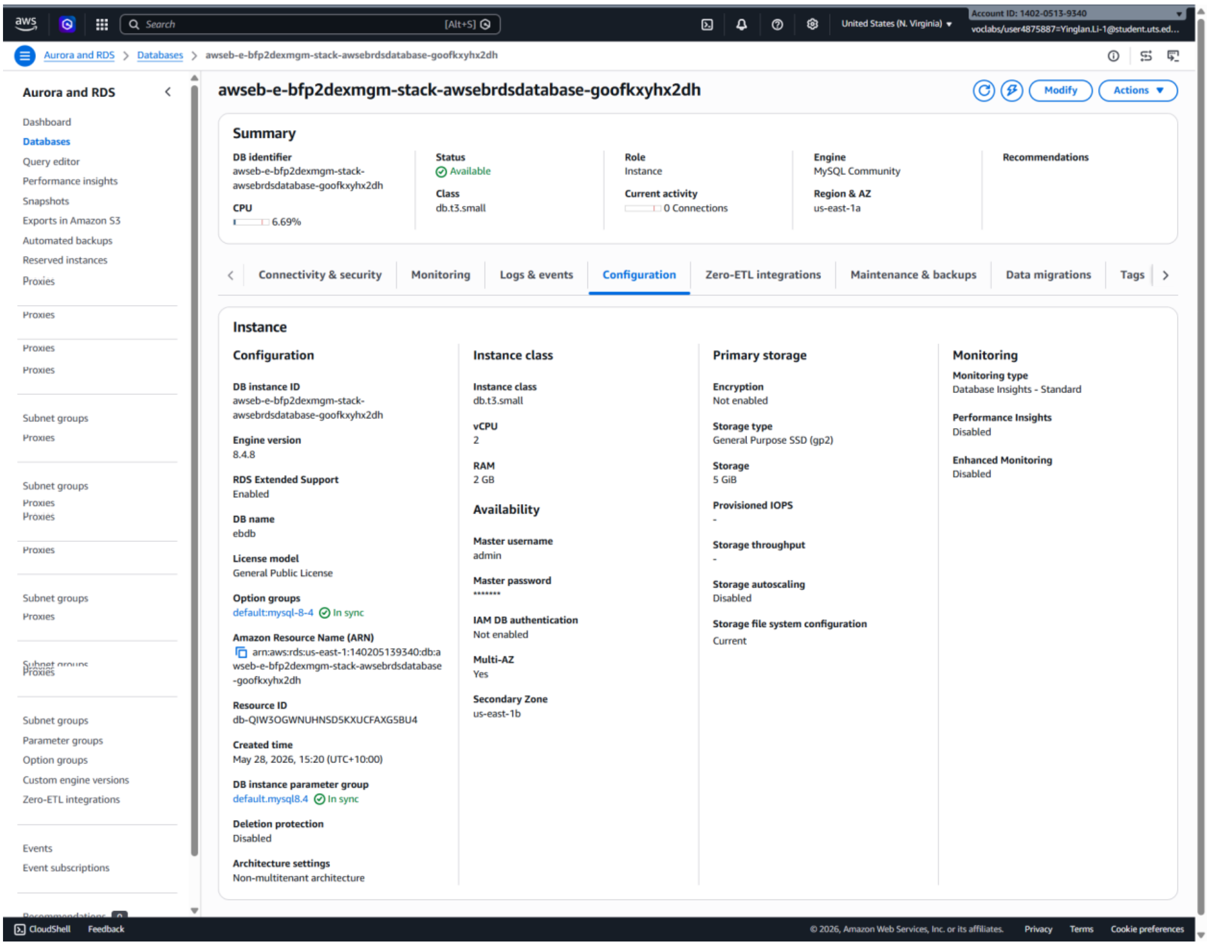
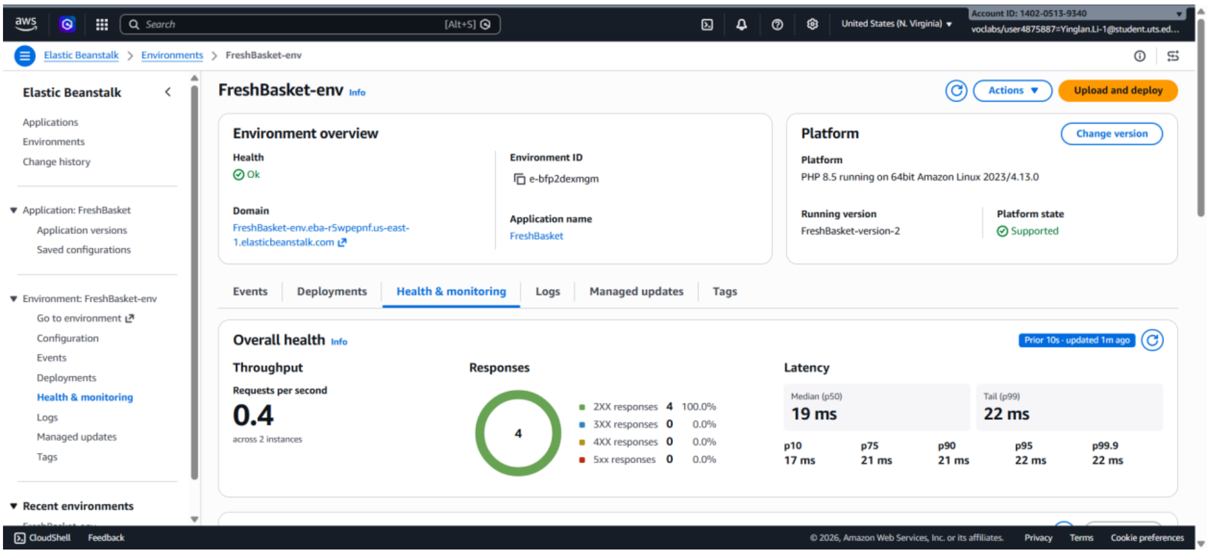


Figure 6. RDS multi-AZ confirmation

This screenshot shows the Amazon RDS MySQL database with Multi-AZ enabled, confirming that the database tier supports high availability and failover across Availability Zones.



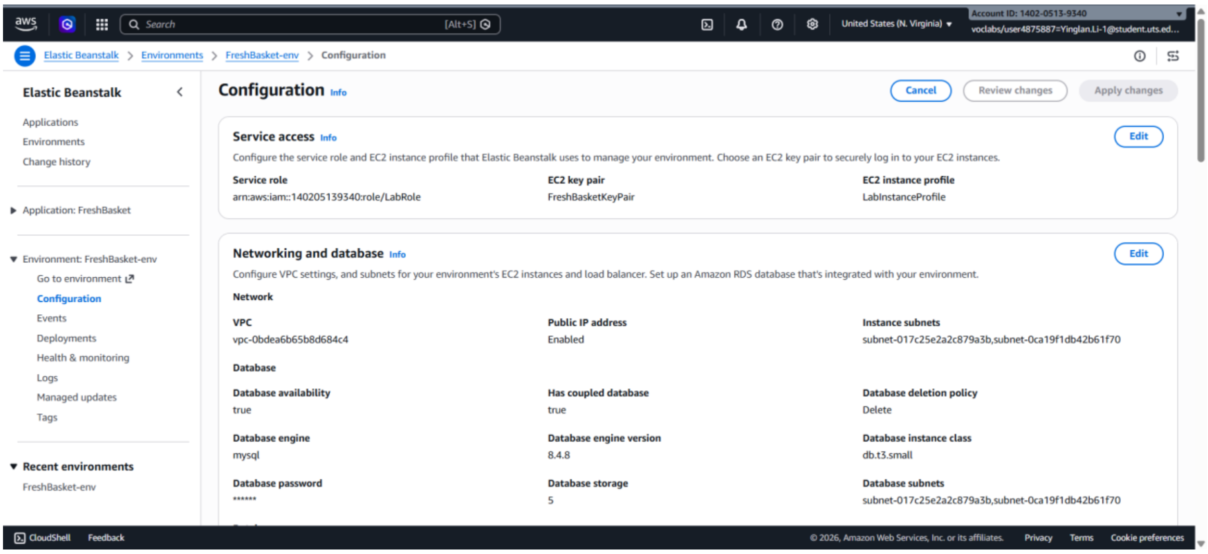


Figure 7. Elastic Beanstalk environment dashboard

This screenshot shows the Elastic Beanstalk environment health status, environment URL, using the LabRole and LabInstanceProfile settings.

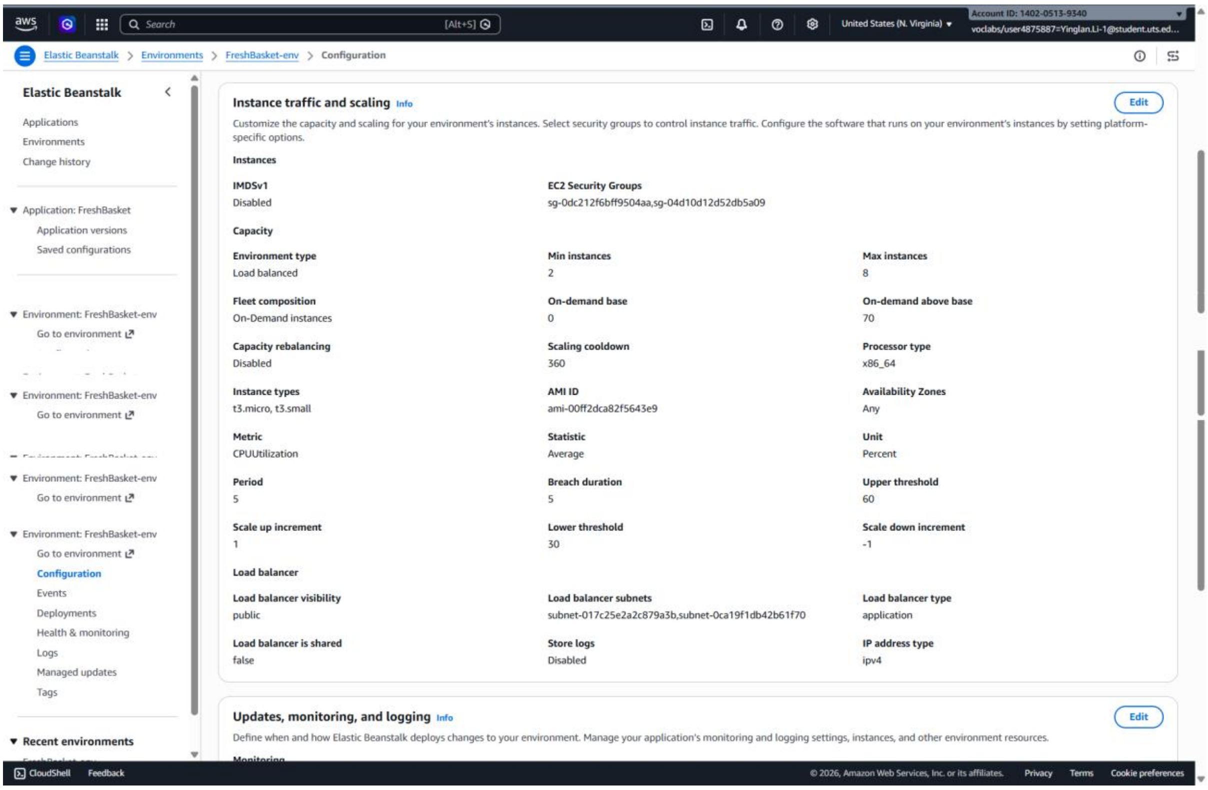


Figure 8. Auto Scaling configuration

This screenshot shows the Auto Scaling configuration with a minimum of 2 instances, a maximum of 8 instances, and CPU utilisation scaling triggers at 60% and 30%.

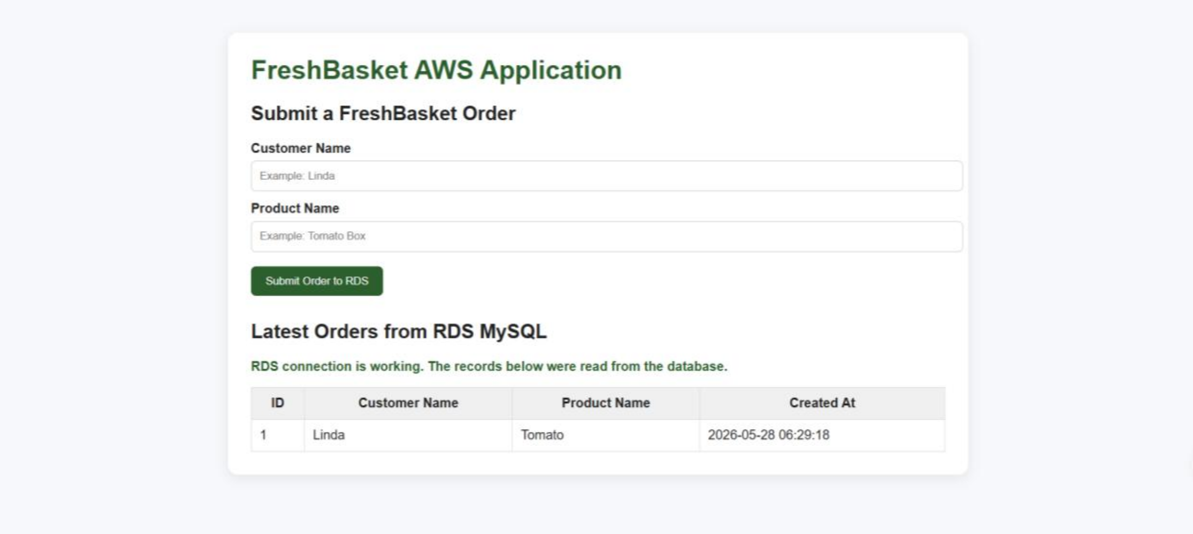


Figure 9. Application running end-to-end with RDS connectivity

This screenshot shows the FreshBasket web application running through the Elastic Beanstalk URL and displaying data read from or written to the RDS MySQL database.

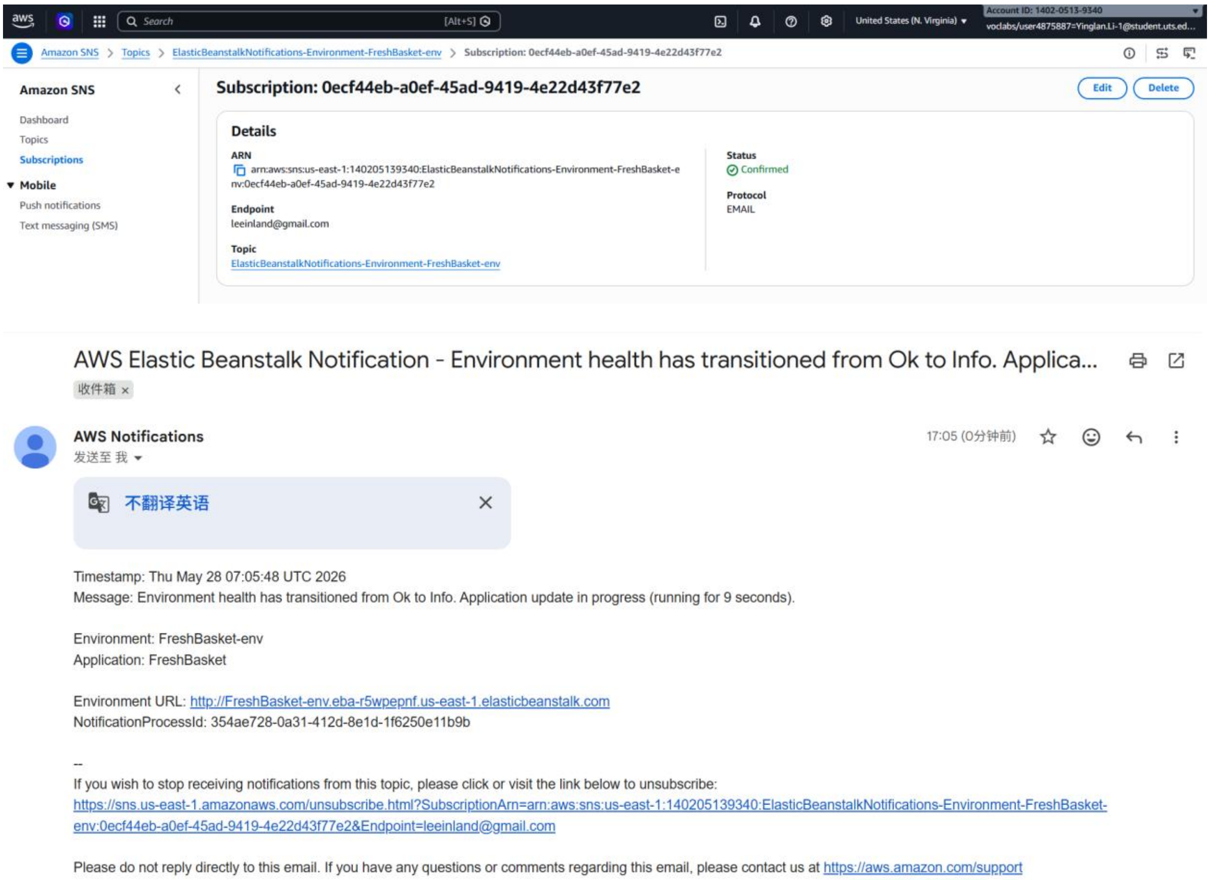


Figure 10. Email notification

This screenshot above shows the SNS email subscription confirmed for Elastic Beanstalk environment notifications. And an Elastic Beanstalk event notification email received through SNS after a new application deployment event.